

# Lab Exercise: High-Frequency Arbitrage in Fragmented Markets

## 1. Context: The Fragmented Market

In modern European equity markets, liquidity is **fragmented**. The same stock (ISIN) trades simultaneously on the primary exchange (BME) and various Multilateral Trading Facilities (MTFs) like CBOE, Turquoise, and Aquis.

Due to this fragmentation, temporary price discrepancies occur. A stock might be offered for sale at €10.00 on Turquoise while a buyer is bidding €10.01 on BME. A High-Frequency Trader (HFT) can profit from this by buying low and selling high instantaneously.

However, these opportunities are fleeting. The "theoretical" profit you see in a snapshot might disappear by the time your order reaches the exchange due to **latency**.

### The Mission

You have been hired as a Quantitative Researcher at a proprietary trading firm. Your boss has given you a dataset of high-resolution market data and asked you to answer three critical questions:

1. **Do arbitrage opportunities still exist in Spanish equities?**
2. **What is the maximum theoretical profit** (assuming 0 latency)?
3. **The "Latency Decay" Curve:** How quickly does this profit vanish as our trading system gets slower (from 0µs to 100ms)?

## 2. Data Specifications

You are provided with a DATA\_BIG/ folder containing subfolders for specific trading dates. Inside, you will find three types of compressed CSV files for various instruments.

### File Naming Convention

The naming pattern for all three file types (QTE, STS, TRD) is:

<type>\_<session>\_<isin>\_<ticker>\_<mic>\_<part>.csv.gz

| Field   | Description               |
|---------|---------------------------|
| type    | QTE, TRD, or STS          |
| session | Trading date (YYYY-MM-DD) |

|               |                                                                                                |
|---------------|------------------------------------------------------------------------------------------------|
| <b>isin</b>   | Cross-venue <b>ISIN</b> (International Securities Identification Number)                       |
| <b>ticker</b> | Venue-specific trading symbol (distinguishes multiple books for the same ISIN on the same MIC) |
| <b>mic</b>    | Market Identifier Code (MIC, e.g., XMAD)                                                       |
| <b>part</b>   | Integer part number. Assume it is always 1 for simplicity.                                     |

### Order Book Identity and Join Key

A single **order book identity** is defined by the tuple:

(session, isin, mic, ticker)

This identity is the **key used to join** corresponding QTE, TRD, and STS data belonging to the same book.

### File Types

1. **QTE (Quotes/Snapshots):** Represents the state of the order book (up to 10 levels deep).
  - epoch: Timestamp in microseconds (UTC).
  - px\_bid\_0, px\_ask\_0: Best Bid and Best Ask prices.
  - qty\_bid\_0, qty\_ask\_0: Available volume at the best price.
  - *Note: Columns exist for levels 0-9.*
2. **STS (Trading Status):** Updates on the market phase (e.g., Open, Auction, Closed).
  - epoch: Timestamp.
  - market\_trading\_status: An integer code representing the state.
3. **TRD (Trades):** Represents the transactions. Not needed for this exercise.

**Note:** you can also find a **DATA\_SMALL** folder that you can use to test quickly without needing to run the simulation over all the data.

### CRITICAL: Vendor Data Definitions

Real-world financial data is rarely clean. The data vendor has provided the following specifications. **Ignoring these will result in massive errors in your P&L calculation.**

#### A. "Magic Numbers" (Special Price Codes)

The vendor uses specific high-value constants to indicate non-tradable states (e.g., Market Orders during auctions). **These are NOT real prices.** If you treat 999,999 as a valid bid, your algorithm will assume you can sell for a million euros.

| Value      | Meaning                | Action Required |
|------------|------------------------|-----------------|
| 666666.666 | Unquoted/Unknown       | Discard         |
| 999999.999 | Market Order (At Best) | Discard         |
| 999999.989 | At Open Order          | Discard         |
| 999999.988 | At Close Order         | Discard         |
| 999999.979 | Pegged Order           | Discard         |
| 999999.123 | Unquoted/Unknown       | Discard         |

## B. Market Status Codes

You can only trade when the market is in **Continuous Trading**. If you trade during an Auction, a Halt, or Pre-Open, your order will not execute immediately. A snapshot is only valid/addressable if the STS for that venue is one of these codes:

| Venue     | Continuous Trading Code |
|-----------|-------------------------|
| AQUIS     | 5308427                 |
| BME       | 5832713, 5832756        |
| CBOE      | 12255233                |
| TURQUOISE | 7608181                 |

## 3. Implementation Guide

You are encouraged to use AI tools (ChatGPT, Claude, etc.) to generate the Python/Pandas code. However, **you** are responsible for the logic and the financial validity of the results.

### Step 1: Data Ingestion & Cleaning

- Write a function to load the QTE and STS files for a given ISIN.
- **Task:** Ensure you are using only valid prices
- **Task:** Ensure you are only looking at addressable orderbooks

### Step 2: Create the "Consolidated Tape"

- To detect arbitrage, you need to compare prices across venues *at the exact same time*.
- **Task:** Create a single DataFrame where the index is the timestamp, and the columns represent the Best Bid and Best Ask for **every** venue (BME, XMAD, CBOE, etc.).

### Step 3: Signal Generation

- **Arbitrage Condition:** An opportunity exists when Global Max Bid > Global Min Ask.
- **Profit Calc:**  $(\text{Max Bid} - \text{Min Ask}) * \text{Min}(\text{BidQty}, \text{AskQty})$ .
- **Rising Edge:** In a simulation, if an opportunity persists for 1 second (1000 snapshots), you can only trade it *once* (the first time it appears). Ensure you aren't "double counting" the same opportunity. If the opportunity vanishes and quickly reappears you can count it as a new opportunity for simplification.
- **Simplification:** only look at opportunities between Global Max Bid and Global Min Ask. There might be others at the second or third price levels of the orderbook, but let's make it simple and use only the best Bid Ask of each trading venue.

#### Step 4: The "Time Machine" (Latency Simulation)

- In reality, if you see a price at time  $T$ , you cannot trade until  $T + \Delta$ .
- **Task:** Simulate execution latencies of [0, 100, 500, 1000, 2000, 3000, 4000, 5000, 10000, 15000, 20000, 30000, 50000, 100000] microseconds
- **Method:** If a signal is detected at  $T$ , look up what the profit *actually is* at  $T + \text{Latency}$  in your dataframe.

## 4. Deliverables & Evaluation

Submit a Jupyter Notebook containing your code and the following analysis:

1. **The "Money Table":** A summary table showing the Total Realized Profit for all processed ISINs at each latency level.
2. **The Decay Chart:** A line chart visualizing how Total Profit (Y-axis) decays as Latency (X-axis) increases.
3. **Top Opportunities:** A list of the Top 5 most profitable ISINs (at 0 latency). **Sanity check these results**—do they look real?

#### Grading Rubric (Max 10 Points)

- **5-6 Points (Baseline):** The code runs, correctly calculates the consolidated tape, identifies Bid > Ask opportunities, and estimates theoretical (0 latency) profit.
- **7-8 Points (Robust):** The simulation accurately models latency (using strict time-lookups) and strictly adheres to the vendor's data quality specs
- **9-10 Points (Expert):** You demonstrate deep understanding of market microstructure. You handle **Market Status** correctly to avoid fake signals, identify anomalies in the instrument list, and handle edge cases around Market Open/Close.

Deadline 09<sup>th</sup> of December 23:59 CET

Submitted to [francisco.merlos@six-group.com](mailto:francisco.merlos@six-group.com) with the title: "Arbitrage study in BME | Your name". Example: "Arbitrage study in BME | Francisco Merlos".

